

Tools To Introduce INTELLIGENCE Into Your MACHINES

SHANOSH KUMAR

idealCircuit (design and simulation)

Engineers often search for light-weight software that has an intuitive interface, fewer components, and models for ease of design and saving time. When designing a faster prototype within limited time, idealCircuit design tool allows you to design schematics and simulate these for specialised applications. While it was initially used for designing power supplies, today it works along NLS analogue electronic circuit simulator on ideal and piecewise-linear components. This software lets you evaluate and simulate the exact same schematics you see in textbooks.

DesignLab (design and simulation)

DesignLab is equipped with functionalities, and you can see your projects lift off the ground in no time. It works on Windows and Linux operating systems and is supercharged by adding circuits. This helps you push your projects into production through faster design creation. Like all integrated development environments (IDEs), Papilio DesignLab lets you develop a hardware description language by typing the code into the IDE, which, in turn, generates a file that is loaded into the field-programmable gate array. Papilio IDE includes a text editor, libraries, revision control, syntax highlighting and a compiler. The unique selling point is the drag-and-drop interface for designing field-programmable gate arrays using Arduino IDE. The tool is limited to Papilio boards only.

Some popular resources

JEdit. Get used to one of the best text editors designed for Java programmers. This software supports Unicode, text encoding sniffer, customisable syntax colouring, customisable key bindings and object embedding such as an image or a movie file. Like other text editors, it has better appeal while working with Java and XML due to the powerful search engine for regular expressions, syntax highlighting and auto-indentation.

Pinn. Pinn makes it easy to select and install operating systems for Raspberry Pi. It comes with the advantage of not having to manually launch an image of the operating system from the SD card and going through the whole installation process.

Cyberfox Portable. This portable browser intelligently uses 64-bit system resources, giving you a faster running seamless Internet surfing experience.

PCB123 (PCB)

PCB123 helps you take advantage of all enhanced user controls for faster rendering, smoother drag and drop, more zoom control and unified control panels while designing PCB boards. The upgraded version included in this month's DVD lets you access a fully-defined set of parts library, with over 500,000 parts with schematic symbols, footprints, manufacturer and digi-key ordering information and links to data sheets.

ECAVA IGX (SCADA)

IGX SCADA is your answer to how modern SCADA software works. Watching over your factory floor round the clock, this software intelligently tracks and sends out emails and SMSes based on hierarchy and access levels. The software runs and links smoothly to any modern Web browser to provide a bird's eye view

on supervisory control. Its reporting system is very real-time, and presentation of production or monitoring data is built in a comprehensive mathematic functionality to increase accuracy of readings. It decreases downtime and increases productivity with multi-level role based supervision and trackable event based audit trail.

OpenCog (utility)

In the race to create artificial intelligence, OpenCog finds itself as the most unique and ambitious open source project that aims to impart intelligence to machines. You can join the community and develop codes and software applications that are at par with or beyond human intelligence. Prime areas of focus are the practical applications used for interacting with humans such as natural language processing and speech.

CoreOS (utility)

This utility software is created as an operating system to be operated in large clusters of servers-supporting Cloud. It is based on open source Linux containers and supports container systems out-of-the-box. CoreOS has been designed to produce, maintain and utilise open source software for Linux containers and distributed systems. If you are a computer engineer and researcher who likes programming and providing solutions on Cloud based applications while you make use of emulator, debugger and flash programming utilities, this software is for you. **EFY**

Shanosh Kumar is technology journalist at EFY. He is BCA from Bangalore University and MBA from Christ University, Bengaluru